

Project Proposal-Group04

Project Name:

TalentVerify AI – Intelligent Candidate Screening and Professional Profile Verification System

1. Overview

Intelligent Resume Skill Analysis and Candidate Grading System is an AI-powered web application that automates the resume screening process for recruiters. Candidates upload their resumes (PDF or DOCX), and the system extracts key information such as skills, education, work experience, certifications, and projects.

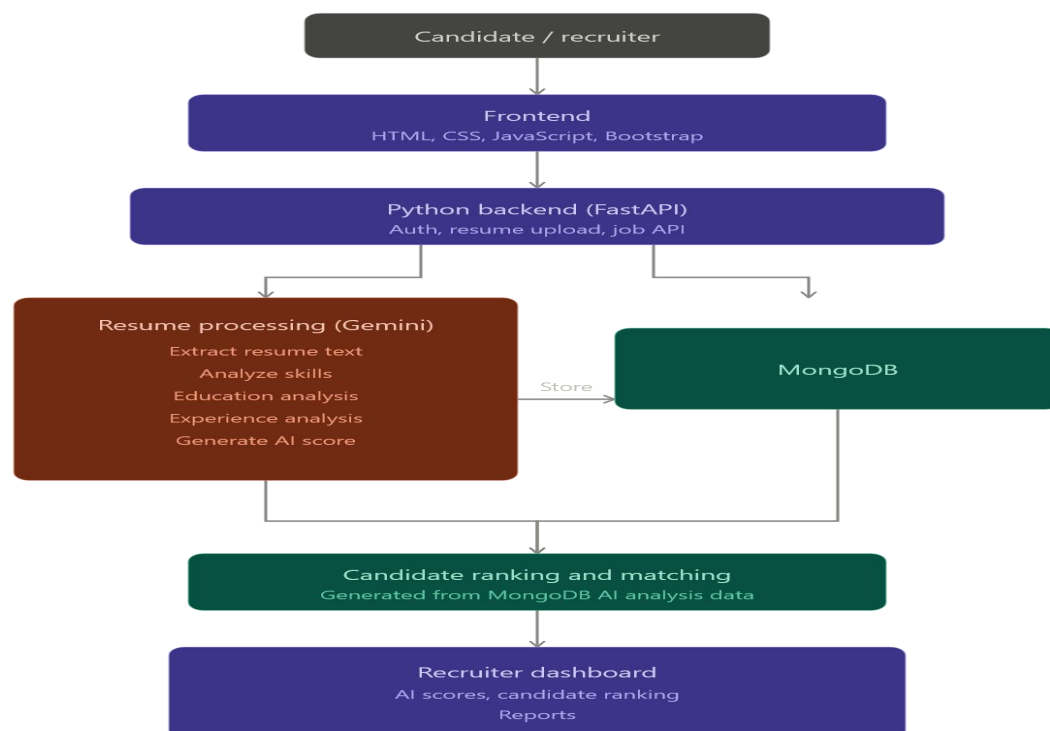
The system uses the Google Gemini API to analyse the resume and compare it with a job description. It then generates a candidate match score, identifies missing skills, and provides recommendations for improvement.

This solution helps recruiters save time, improve candidate evaluation, and provide applicants with useful feedback to enhance their employability.

2. Challenges Intended to Solve Using AI Integration

- Manual resume screening is time-consuming.
- Recruiters receive hundreds of applications for a single vacancy.
- Candidates often don't know which skills are missing for a job.
- Comparing multiple resumes consistently is difficult.
- Traditional keyword-based screening may overlook qualified candidates.
- HR teams need faster and more consistent hiring decisions.

3. System Architecture Diagram



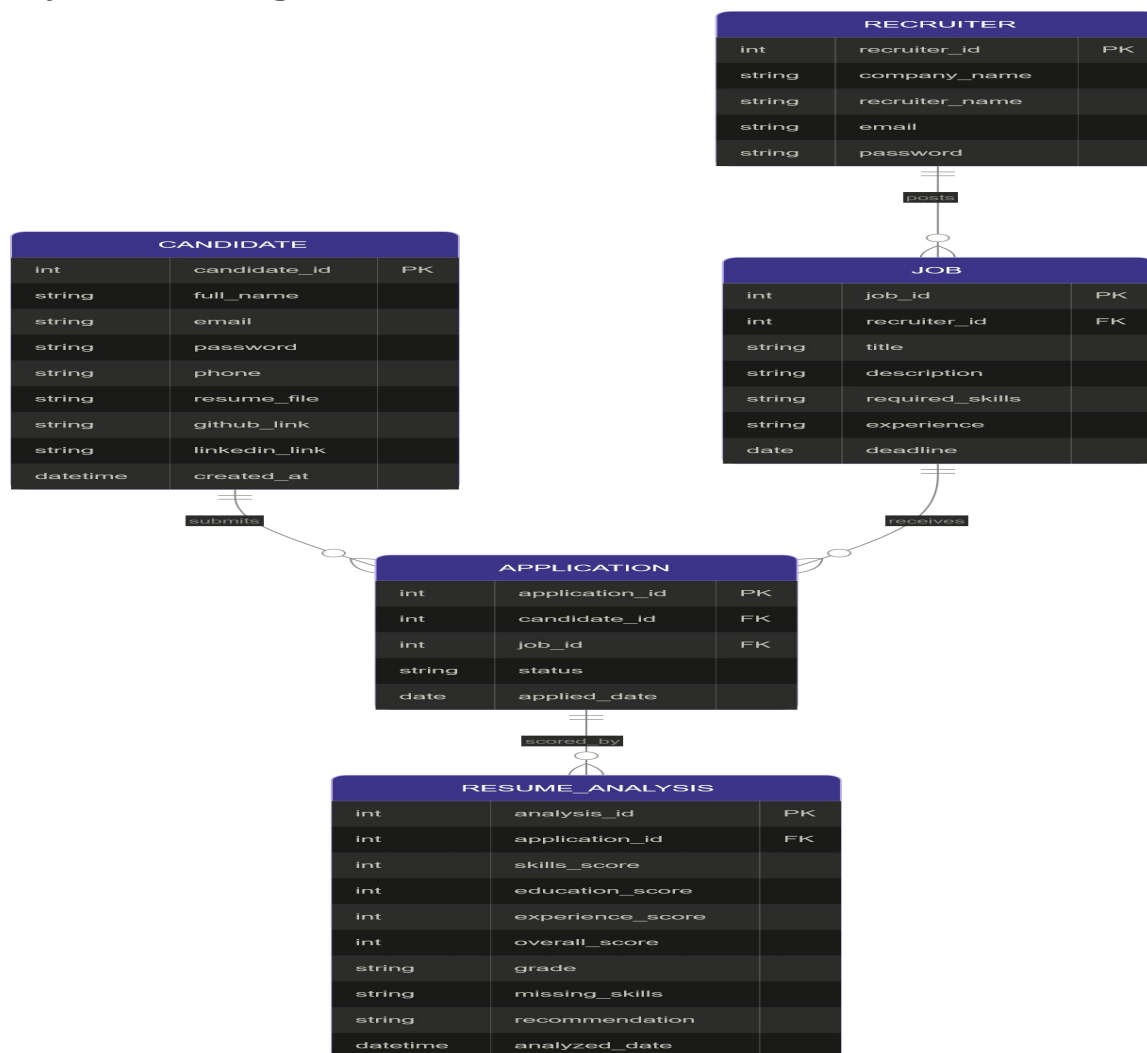
4. Functional Requirements

- Candidate Functions, Recruiter Functions, AI Functions

Non-Functional Requirements

-Performance, Security, Privacy, Fairness and Explainability, Maintainability

5.System ER Diagram



6.Technology Details

Frontend -- HTML5 , CSS3 ,JavaScript ,Bootstrap

Backend -- Python, Fast API , REST API

AI Integration -- Google Gemini API

Resume Processing -- Google Gemini API

Database -- *MongoDB*

7.Conclusion

The Intelligent Resume Skill Analysis and Candidate Grading System provides an efficient and AI-powered solution for automating the resume screening process. By integrating the Google Gemini API, the system analyses resumes, compares them with job descriptions, generates candidate scores and grades, and provides recommendations for improvement. This helps recruiters make faster and more consistent hiring decisions while enabling candidates to identify skill gaps and improve their employability.

8.GitHub Repository Link

<https://github.com/Erangaisuru22/talentverify-ai>